

Far Red Postables: Ultra-low-cost Pupillometry

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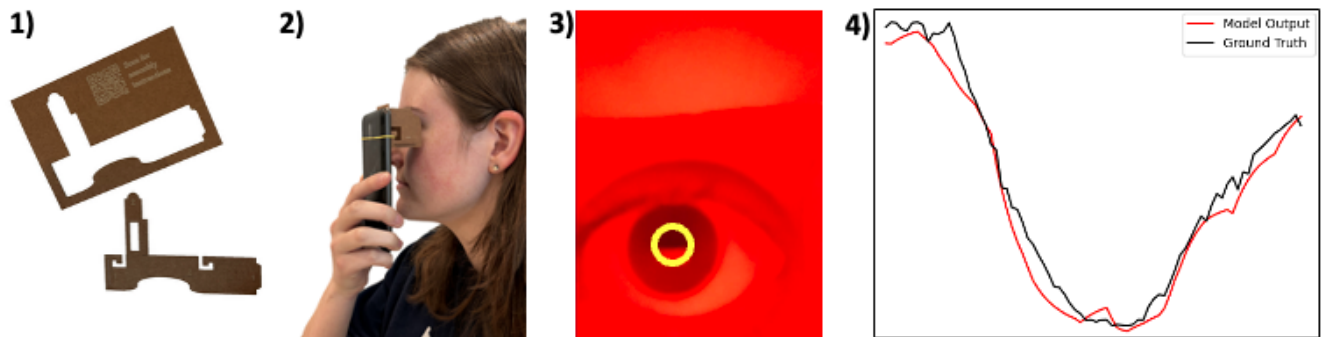


Figure 1: Far Red Postable Enables Ubiquitous Pupillometry. 1) Postable in flat form, prior to construction. 2) Constructed postable in use with smartphone. 3) Screenshot of pupil during measurement, where pupil is located in yellow circle. 4) Sample output of pupil relative size during pupillary light reflex test.

Abstract

Pupillometry is an important measurement in neurological exams and has been shown to have increased utility in at-home screenings, especially for Traumatic Brain Injury. However, the high cost of clinical grade pupillometers limits their accessibility. Recent work has explored smartphone-based pupillometry, but for various accessibility reasons, current solutions are not truly ubiquitous. This demo paper introduces the Far Red Postable system, an ultra-low-cost, racially fair, smartphone-based pupillometer. We demonstrate that the Far Red light spectrum can be effectively used without any specialized electronics or 3D-printed attachments. Using a cheap red light filter, paper-based attachment, and a rubber band, we can turn a smartphone into a truly ubiquitous pupillometer.

CCS Concepts

• **Human-centered computing** → Ubiquitous and mobile computing; • **Applied computing** → Life and medical sciences.

Keywords

Pupillometry, Smartphone, User-Assembled



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1 Introduction

Pupillometry is the study of pupil size and reactivity to stimuli. As the pupil's response to stimuli sheds light into neurological functions, pupillometry is an important measurement in clinical neurological exams [10]. Clinically, pupillometry measurements can aid in diagnosing and screening for neurological diseases including Alzheimer's Disease, Parkinson's Disease, Schizophrenia, Mild Cognitive Impairment, and Traumatic Brain Injury [4, 5, 7, 8, 12, 16, 18]. A large body of work has focused on more in-the-wild pupillometry measurements and its usefulness in screening for Traumatic Brain Injury, in other words, concussions [6, 9]. However, there is no clear solution for an accessible pupillometer to use in these in-the-wild screenings. Here, we outline the current solution space to provide context for the novelty of this presented work.

1.1 Clinical Pupillometers

Clinical pupillometers cost multiple-thousands of dollars due to their precise instrumentation and utilization of infrared light to achieve contrast between the pupil and iris. While favorable for

clinical diagnosing, this high price tag makes the clinical solutions impracticable for ubiquitous concussion screenings.

1.2 Smartphone-based Pupillometers (With No Attachments)

The ubiquitous nature of smartphones makes them an ideal platform for performing in-the-wild pupillometry. Many researchers have explored this route [1, 9, 11, 13]. Smartphones with an on-board infrared camera can be relatively accessible for performing pupillometry measurements [1]. However, most smartphones do not contain infrared cameras, and thus this solution is not truly ubiquitous as it relies on unique smartphone models [2]. In contrast, it is a fair assumption that all smartphones can support RGB camera measurements. As such, we must focus on smartphone-based solutions that rely on the RGB camera. Most prior works can achieve pupillometry measurements on people with light iris colors, where there is high contrast between the pupil color and iris color [9, 11, 13]. However, with a vast majority of the global population having brown or other dark-colored irises, this is not a fair solution.

1.3 Smartphone-based Pupillometers (Racially Fair)

Prior work by Barry et al. has demonstrated the ability of a red light filter to be used in conjunction with RGB smartphone cameras to enable not only pupillometry measurements on light-colored irises, but dark-colored irises as well [3]. This work takes advantage of the “Far Red” light spectrum, in which only the red channel of light is most active, and there is higher contrast between pure black pupils and dark, but not pure black, irises. However, this work relies on an elaborate attachment to be used in conjunction with the phone, including a custom printed circuit board with additional LEDs, a thick red light glass filter, and specific 3D-printed mounting. So, while more accessible than pure smartphone-based solutions in its ability to screen all people, and more accessible than costly clinical solutions, this work still is unrealistically ubiquitous.

1.4 Our Pupillometer (Racially Fair, Accessible)

We introduce the first racially fair, passive, ultra-low-cost pupillometer system that is truly ubiquitous. We enable the same Far Red pupillometry phenomenon, but without the reliance on a niche, active system attachment. Instead, we only require a paper attachment, a cheap low-pass filter, and a rubber band – in conjunction with the downloadable smartphone app. The attachment system, shown in Figure 2, has no electronic components and is fully assembled by users. This makes our solution truly ubiquitous: it can even be deployed through the postal service to users, just as easy as getting your daily newspaper. Just get your device, assemble, and take your measurement on your smartphone.

2 Far Red Postable Pupillometer

2.1 Concept and Prototype

The arguably most important part of the original system is the ideal low-pass red light filter. However, as this is glass, it is important for our solution to find an alternative to this filter that can still enable



Figure 2: Far Red Postable System. Postable attachment with cheap red light filter connected to smartphone with rubber band. Filter rests over front-facing camera.

the same Far Red phenomena. We acquired and tested three different cheap low-pass red light filters and compared them to the ideal filter from the solution in prior work [15]. Figure 3 demonstrates the performance of each of these filters across the light spectrum. As can be seen, selecting the curve that is as close to the ideal curve (where light is cut-off before the Far Red spectrum) as possible will give us the closest results. Therefore, we designed our prototype with the spy filter as it had the closest performance. This filter costs approximately 25 cents for each attachment system and is thus comparable to the price of a postage stamp [17].

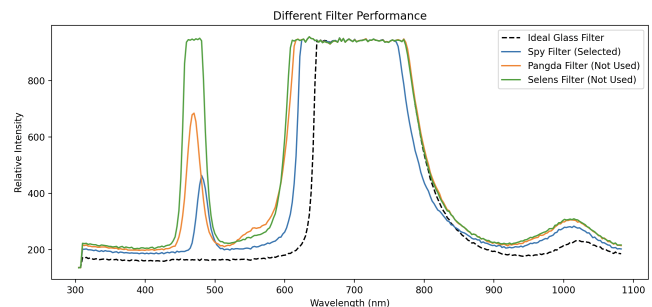


Figure 3: Spectrogram of Three Plastic Red Light Filters Compared to Ideal Glass Filter.

To facilitate a pupillometer measurement, hold the filter in place in front of the smartphone camera, and aid in blocking out ambient light that may affect pupillometry screenings, we designed the necessary attachment system. Instead of using 3D-printed material, we designed the system out of a cardstock paper and laser cut the

design into it. As such, a user can ‘punch out’ the system along the proliferation and fold the system together according to the app instructions. Figure 4 demonstrates the different stages of assembly of the Postable device and how the system attaches to the smartphone with a rubber band. Note that the plastic red light filter is not included in these images (so the smartphone front-facing camera can be viewed in the gap where the filter would be attached). In final deployment, the filter will come pre-attached to the postcard, so the user does not have to assemble that part themselves.

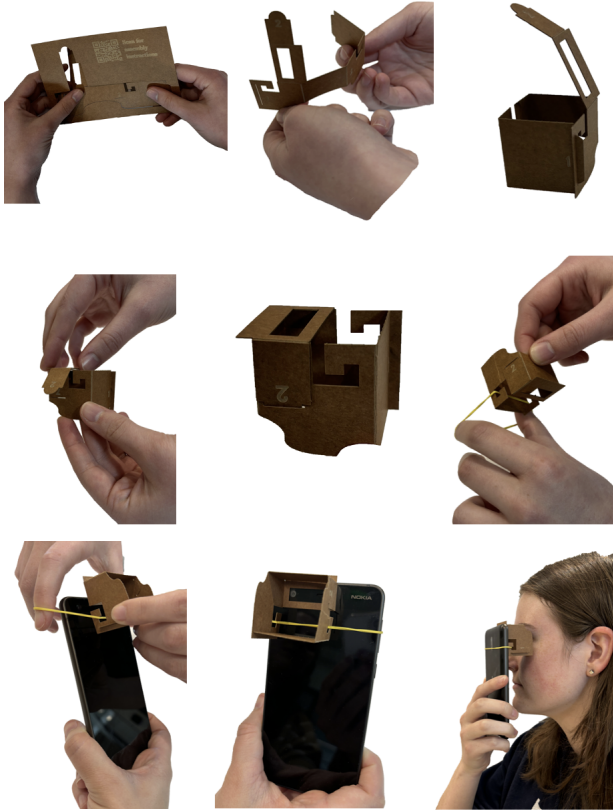


Figure 4: Assembly Steps for the Far Red Postable (note no filter shown). (Top left) User punches proliferated edges out of template. (Top middle) User folds along the creases and inserts tab 1 into slot 1. (Top right) Postable after step 1 complete. (Middle left) User folds along the crease to insert tab 2 into slot 2. (Middle middle) Postable after fully assembled. (Middle right) User inserts rubber band into the cut-outs. (Bottom left) User wraps rubber band around smartphone. (Bottom middle) Postable appropriately secured over smartphone front-facing camera. (Bottom right) User holds smartphone and Postable to eye socket for pupillometry testing.

Once the user has assembled their system, they then follow the on-app instructions for a given pupillometry measurement (for example, a pupillary light reflex (PLR) test). Figure 5 shows screenshots of the application the user follows for building and testing instructions. The video of their pupil (collected via the smartphone

camera) is then fed into a fine-tuned YOLO segmentation model pre-trained on locating the pupil, and a pupil diameter is extracted at every video frame to get their PLR curve and associated concussion screening assessment. It is not the scope of this work to provide screening results or clinical interpretations. Instead, we show that we can achieve a comparable PLR curve as a clinical pupillometer, thus demonstrating the measurement system is equivalent.

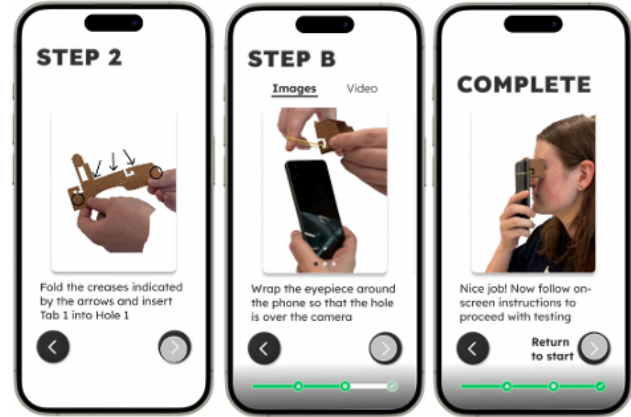


Figure 5: Smartphone Application Instructions for the Far Red Postable. (Left) Screenshot of one instruction the user sees for assembly. (Middle) Screenshot of one instruction the user sees for attachment to smartphone. (Right) Final screen user sees before measurement begins.

2.2 Testing and Validation

A full-scale, journal-ready validation and analysis of the system is still in progress. However, we can showcase the intermediary results in demonstration of the working system. Figure 6 demonstrates how the output video from the ML model using the Far Red Postable system can identify the pupil. Specifically, it is clear to see that the red light filter vastly improves one’s ability to locate the pupil of a sample dark-colored iris, compared to a normal RGB snapshot. As such, the model can segment the image to extract the pupil in each frame of the PLR output video. Note that with a light-colored iris, the pupil is more easily identified, and as such a dark iris example is included to demonstrate that the system enables this difficult detection.

Further, we can compare the results of the Far Red Postable system directly with an FDA-approved clinical pupillometer, the NeuroOptics PLR-3000 [14], to validate that the correct pupil size is detected. By holding the system up to one eye and applying the stimulus, while the NeuroOptics device is held up to the participant’s other eye, we can collect a ground truth measurement to compare our Postable system’s measurement with. Figure 7 shows one participant’s results where the filtered output from the Postable system is compared directly to the NeuroOptics output. Note that there is no time synchronization between the two devices, and as such, the ground truth curve is manually shifted in time to visualize the shape comparison. As can be seen, the system follows a comparable path, demonstrating that our prototype works successfully.

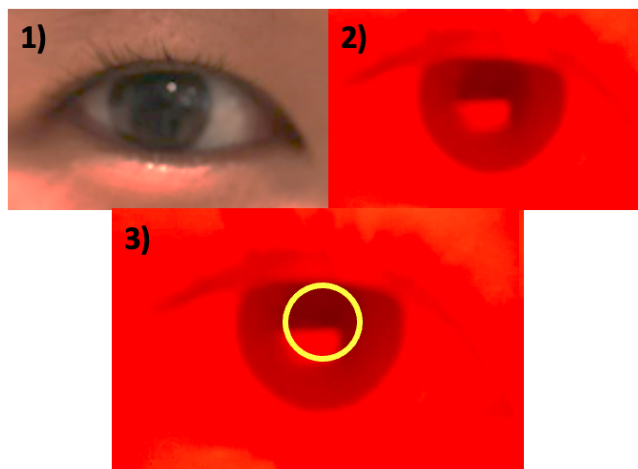


Figure 6: Pupil detected from system's output PLR test video for one participant. 1) RGB captured image of participant's eye (note this image has been brightened in post-processing). 2) Screenshot of video taken by smartphone during PLR test with the system. 3) Screenshot of model output video where pupil is identified.

Again, a more robust analysis is currently in progress, but these initial results are included to demonstrate that the system currently works.

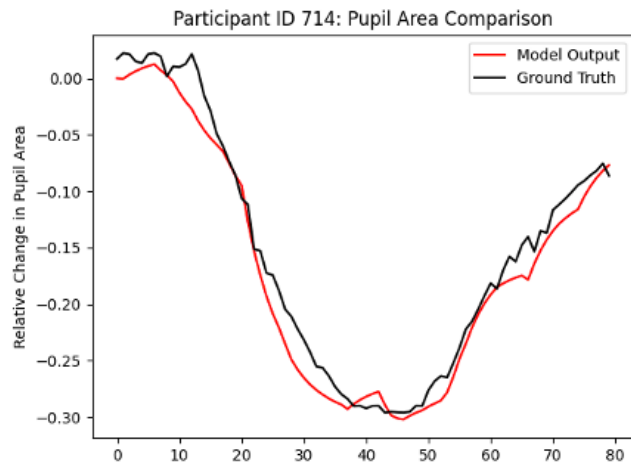


Figure 7: Relative Pupil Change during PLR Test Over Video Frame Arbitrary Time Steps. The Far Red Postable system's pupil size is shown in red, and ground truth measured using the NeuroOptics PLR-3000 is shown in black.

3 Discussion

The presented results demonstrate the ability of the Far Red Postable system to facilitate Far Red pupillometry measurements without specialized electronics or an ideal long-pass filter. As stated above, further in-depth testing is required to demonstrate the robustness

of the system. Additionally, it is important to analyze the effect of iris color on the system's performance. As an intermediary result, Figure 6 demonstrates the system's ability to perform on a dark-colored iris. Similarly, Figure 7 presents the analysis on the same eye as depicted in Figure 6. As such, we demonstrate an example of the Far Red Postable system performing adequately on the population of users that will be most difficult to achieve satisfactory results.

While the given work's contribution of expanding access to pupillometry is important, it is necessary to highlight key limitations of the system as well. First, the system is incapable of reporting the absolute pupil size and instead can only measure relative changes in size. While a calibration could be performed to approximate this value, this would rely on too many assumptions of the user's unique operation and facial structure, and thus we determined this to be an inappropriate endeavor.

Next, the model relies on a bright ambient environment to be able to visualize the pupil. In other words, if you do the measurement in a dark room, the system may not be able to report accurate pupil sizes. We believe that this environment requirement is reasonable. However, it is important to note that this does limit the system's ubiquity as a user could not rely on this test in an outdoors, nighttime setting.

Finally, the scope of this work is limited to recreating a clinically similar result in a PLR test. We cannot yet comment on the ability of the Far Red Postable to screen for concussions or Traumatic Brain Injury in-the-wild. Future work is necessary to assess the diagnostic performance of the system.

4 Conclusion

This demonstration presents the first passive, ultra-low-cost solution for enabling racially fair Far Red pupillometry. With a cheap red light filter and a paper-based attachment to facilitate the measurement, we are able to achieve pupil detection and tracking with a similar resolution as a clinical grade pupillometer. Without the reliance on specialized hardware, Far Red Postables make pupillometry measurements ubiquitous.

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